

CASABLANCA ANFA, Morocco

WMO: 601550

Lat: 33.557N

Lon: 7.660W

Elev: 62

StdP: 100.58

Time Zone: 1.00 (E01)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	7.9	2.9	4.7	10.6	4.3	5.2	11.1	7.5	14.2	6.3	14.6	1.4	200	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	29.5	21.5	27.6	21.9	26.4	21.8	24.0	26.8	23.3	26.0	22.7	25.3	2.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	18.0	25.6	22.3	17.1	24.8	22.0	16.8	24.4	72.3	26.8	69.7	26.0	67.5	25.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	4.9	4.9	34.6	1.7	3.2	3.7	36.9	2.7	38.8	1.7	40.6	0.5	42.9		
(5)			3.6	25.2	1.9	1.2	2.3	26.1	1.2	26.8	0.2	27.5	-1.2	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	13.5	14.0	15.8	17.0	19.0	21.7	23.1	23.5	22.5	20.6	17.0	14.7		
	DBStd	4.05	2.07	2.11	2.32	1.96	2.02	1.75	1.53	1.32	1.62	2.05	2.59	2.10		
	HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	599	151	122	86	49	15	0	0	0	0	3	56	116		
	CDD10.0	3124	109	113	179	209	279	350	406	419	377	329	209	145		
	CDD18.3	680	1	1	7	9	36	101	147	161	127	74	16	2		
	CDH23.3	2121	3	3	42	37	87	256	494	589	375	184	46	5		
(13)	CDH26.7	288	1	1	12	8	18	51	76	57	28	29	8	1		
(14)	Wind	WSAvg	2.3	1.8	2.1	2.3	2.6	2.6	2.5	2.3	2.2	2.0	1.9	2.0		
(15) Precipitation	PrecAvg	422	66	54	47	37	18	4	1	0	7	40	71	75		
	PrecMax	806	314	214	189	156	100	26	8	5	45	133	253	378		
	PrecMin	148	0	0	0	0	0	0	0	0	0	0	1	0		
	PrecStd	142	60	46	40	29	22	6	2	1	10	38	53	69		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.1	27.7	27.7	29.7	33.5	33.1	30.9	29.4	29.9	27.8	23.2		
		MCWB	15.5	16.4	18.1	18.1	19.6	22.3	23.4	23.3	23.4	18.4	19.2	15.8		
	2%	DB	19.7	20.4	23.4	23.8	25.7	27.2	28.4	27.9	27.0	26.8	24.1	20.8		
		MCWB	15.2	16.1	16.8	17.8	19.3	21.8	22.7	23.2	22.4	20.0	18.7	16.5		
	5%	DB	18.1	18.9	21.1	21.8	23.5	25.8	26.8	26.8	26.0	25.1	22.2	19.3		
		MCWB	15.0	15.6	16.6	17.3	19.1	21.3	22.6	23.0	21.8	20.5	18.3	15.9		
	10%	DB	17.1	17.8	19.8	20.2	22.2	24.4	25.7	25.9	25.2	24.0	21.1	18.5		
		MCWB	14.4	15.0	16.2	16.7	18.5	20.8	22.1	22.4	21.4	20.2	17.7	15.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	18.2	19.2	20.1	21.3	23.7	24.6	24.9	24.6	22.8	21.5	18.6		
		MCDB	19.7	21.2	23.9	23.7	25.5	28.0	28.4	27.8	27.5	25.2	24.0	20.8		
	2%	WB	16.2	16.9	18.0	18.7	20.4	22.6	23.7	24.1	23.4	22.0	20.0	17.5		
		MCDB	18.4	19.0	21.5	22.2	23.8	26.1	26.7	26.6	25.7	24.3	22.6	19.7		
	5%	WB	15.5	16.1	17.3	17.8	19.6	21.9	23.0	23.5	22.6	21.4	19.0	16.7		
		MCDB	17.6	18.2	20.3	20.6	22.7	24.9	25.9	25.8	24.9	23.7	21.5	18.7		
	10%	WB	14.8	15.3	16.5	17.2	19.0	21.2	22.5	22.8	22.1	20.8	18.1	15.9		
		MCDB	16.9	17.3	19.2	19.8	21.8	23.9	25.2	25.1	24.4	23.0	20.7	17.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	7.5	7.6	6.6	6.0	5.2	5.0	5.0	5.6	6.9	7.5	7.5		
		MCDBR	8.6	8.9	10.4	8.8	8.1	7.4	7.0	6.2	6.1	8.1	8.7	8.3		
		MCWBR	5.8	5.8	6.1	5.0	4.2	3.7	3.3	3.2	3.2	4.4	5.4	5.4		
	5% WB	MCDBR	7.9	8.1	8.8	7.4	7.2	6.4	6.1	5.8	5.8	6.9	7.6	7.1		
		MCWBR	5.7	5.6	5.6	4.7	4.1	3.6	3.4	3.3	3.5	4.2	5.0	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.360	0.414	0.409	0.433	0.465	0.507	0.504	0.445	0.407	0.368	0.343		
	taud		2.495	2.407	2.228	2.276	2.225	2.136	2.002	2.031	2.226	2.351	2.443	2.494		
	Ebn at Noon		878	887	859	880	858	827	790	785	818	824	829	845		
	Edh at Noon		86	104	134	134	142	155	176	169	133	109	91	82		
(44) All-Sky Solar Radiation	RadAvg		3.00	3.86	5.01	6.15	6.91	7.46	7.49	6.78	5.60	4.18	3.17	2.75		
	RadStd		0.19	0.33	0.34	0.29	0.36	0.30	0.18	0.19	0.22	0.28	0.22	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	-0.86	+0.93	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page